

TECHNICAL REPORT

Document Number and Edition: EHIVE03-294243812-1089 ED 001

Document Title: Minncare for Sporocidal Resin Sanitization

SIGNATURES

Function	Name, Position, Department	Signature	Date
Prepared by	Kevin Hill-Byrne R&D Associate I BioProcess Tech and Eng		
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Workflow Status: BPD Reports eApproval

Workflow Information

Initiator: Ramos, Manny**Document:** [EHIVE03-294243812-1089 ED 001](#)**Started:** 2/26/2019 3:50 PM**Status:** Approved**Last run:** 2/26/2019 9:55 PM

Workflow Visualization

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Tasks

This workflow created the following tasks. You can also view them in [Tasks](#).

☐	Assigned To	Document Number	Due Date	Status	Related Content	Outcome
☒	Hill-Byrne, Kevin	Please approve EHIVE03-294243812-1089 ED 001 	3/1/2019	Completed	EHIVE03-294243812-1089 ED 001	Approved
☒	Arnold, Lindsay	Please approve EHIVE03-294243812-1089 ED 001 	3/1/2019	Completed	EHIVE03-294243812-1089 ED 001	Approved
☒	Brown, Arick	Please approve EHIVE03-294243812-1089 ED 001 	3/1/2019	Completed	EHIVE03-294243812-1089 ED 001	Approved
☒	Rose, Steven	Please approve EHIVE03-294243812-1089 ED 001 	3/1/2019	Completed	EHIVE03-294243812-1089 ED 001	Approved

Workflow History

The workflow recorded these events.

☐	Date Occurred	Event Type	User ID	Description	Outcome
	2/26/2019 3:51 PM	Workflow Initiated	 Ramos, Manny	Approval was started. Participants: Hill-Byrne, Kevin;Arnold, Lindsay;Brown, Arick;Rose, Steven	
	2/26/2019 3:51 PM	Task Created	 Ramos, Manny	Task created for Hill-Byrne, Kevin. Due by: 3/1/2019 12:00:00 AM	
	2/26/2019 3:51 PM	Task Created	 Ramos, Manny	Task created for Arnold, Lindsay. Due by: 3/1/2019 12:00:00 AM	
	2/26/2019 3:51 PM	Task Created	 Ramos, Manny	Task created for Brown, Arick. Due by: 3/1/2019 12:00:00 AM	
	2/26/2019 3:51 PM	Task Created	 Ramos, Manny	Task created for Rose, Steven. Due by: 3/1/2019 12:00:00 AM	
	2/26/2019 3:53 PM	Task Completed	 Brown, Arick	Task assigned to Brown, Arick was approved by Brown, Arick. Comments:	Approved by Brown, Arick
	2/26/2019 3:55 PM	Task Completed	 Hill-Byrne, Kevin	Task assigned to Hill-Byrne, Kevin was approved by Hill-Byrne, Kevin. Comments:	Approved by Hill-Byrne, Kevin
	2/26/2019 4:22 PM	Task Completed	 Arnold, Lindsay	Task assigned to Arnold, Lindsay was approved by Arnold, Lindsay. Comments:	Approved by Arnold, Lindsay
	2/26/2019 9:55 PM	Task Completed	 Rose, Steven	Task assigned to Rose, Steven was approved by Rose, Steven. Comments:	Approved by Rose, Steven
	2/26/2019 9:55 PM	Workflow Completed	 Ramos, Manny	Approval was completed.	Approval on EHIVE03-294243812-1089 ED 001 has successfully completed. All participants have completed their tasks.

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1 EXECUTIVE SUMMARY

This report summarizes the work performed to determine the compatibility of Minncare 1.4% solution with MabSelect SuRe and MabSelect PrismaA resins for use as a sanitizing agent. The data outlined in this report suggests this solution can be statically held in contact with either resin without significantly adverse effects. It is recommended to use this sanitizing agent on an as needed basis, which would be when bioburden data shows typical sanitizing procedures did not kill all spores. Future work will evaluate the effect that dynamic exposure has on MabSelect PrismaA and product quality in the form of a lifetime study.

2 INTRODUCTION

Minncare (Part number 176-01-001) is a cold sterilant manufactured by Mar Cor Purification which consists of Peracetic Acid, Hydrogen Peroxide, and Acetic Acid in an aqueous solution. This sterilant is capable of killing resilient endospores which is otherwise unaffected by 0.1N sodium hydroxide, which is currently used for protein A resin sanitization. Studies performed summarized in NG-1861358 (REPT_ANAL(370477)) assessed 0.1N sodium hydroxide, PAB, and 1.4% Minncare in their ability to kill a MEDI originated *Bacillus* spore and *A. brasiliensis* (16404) in the presence of MabSelect SuRe. Results show Minncare was the only sanitizing agent of the ones tested that was capable of killing both spores (>3 logs reduction) within a reasonable timeframe of 2 hours. However, the effects of Minncare on the resin and product have yet to be studied. This report details the results of experiments conducted to determine how compatible 1.4% Minncare solution is with MabSelect SuRe and MabSelect PrismaA in terms of binding capacity and resulting product quality (eLN ML02604-04). Recommendations on how best to implement this sanitizing solution into new processes will be discussed.

3 MATERIALS AND METHODS

Minncare stock solution was purchased from Mar Cor Purification (Plymouth, MN) which was used to make the 1.4% v/v solutions used in all studies. Minncare 1% Test Strips (Part number 185-40-005) and Minncare Residual Test Strips (Part number 185-40-004) were also purchased from Mar Cor Purification. MEDI4893 UFDF product diluted with 50 mM tris, pH 7.4 was used for dynamic binding capacity studies. MEDI9447 CM was used for the product quality comparison study. The resins used in the dynamic binding capacity studies

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were MabSelect SuRe and MabSelect Prisma. The resin used in the product quality comparison study was MabSelect Prisma.

All chromatography experiments were performed using a GE AKTA Avant with Unicorn Version 7.1 (GE Healthcare, Piscataway, NY, USA). Protein concentration was determined by a protein A affinity HPLC. CHO DNA quantification was performed using PCR in a 96 well plate format. CHO host cell protein quantification was performed using a Gyros CHO-HCP Kit from Gyros Protein Technologies. Aggregate, fragment, and monomer percentages were determined using HPSEC. Residual protein A was determined using sandwich ELISA in a 96 well plate. Insulin quantification was determined using a Gyros insulin assay. Silicone quantification was determined using inductively coupled plasma-mass spectrometry. Pluronic F68 quantification was determined using gas chromatography. Protein charge variance was determined using native imaged capillary isoelectric focusing using an iCE3 instrument from Protein Simple. Quantification of residual Minncare was determined using Minncare Residual Test Strips which are color indicative of hydrogen peroxide concentration.

3.1 Dynamic Binding Capacity Studies Methods

A column was packed with MabSelect SuRe to dimensions of 1.1 x 11.2 cm. Another column was packed with MabSelect Prisma to dimensions of 1.1 x 9.9cm. The columns were equilibrated with 3CV of freshly prepared 1.4% Minncare solution and held statically for a preset amount of time wrapped in foil to prevent light exposure (hold times shown in Table 3.1-1). Hold times are defined as the cumulative amount of time the resin had been exposed to the Minncare solution taking into account all previous runs. For example, Run 5 exposed the column for an additional 46 hours, as the remaining 24 hours of exposure came from the previous 3 runs, giving a sum of 70 total hours. The product used for the dynamic binding capacity studies was MEDI4893 UFDF product diluted to a concentration of approximately 5g/L or approximately 3 g/L with 50 mM tris, pH 7.4 (Table 3.1-1). The product was diluted with equilibration buffer to bring the concentration down to the linear range of UV₂₈₀ measurements. This allows for the UV₂₈₀ trace to be an accurate measure of breakthrough percentage. Both columns were run in the following conditions.:

- 1) Equilibration with 3CV 1.4% Minncare and hold for specified amount of time
- 2) Equilibration with 3CV 50 mM Tris, pH 7.4
- 3) Load diluted MEDI4893 UFDF product to over 10% breakthrough as determined from the UV₂₈₀ chromatogram trace (load concentrations are in Table 1)

- 4) Strip with 3CV 100 mM Acetic Acid
- 5) Caustic Strip with 3CV 0.1N Sodium Hydroxide
- 6) Store with 20% Ethanol

All steps were run at 300 cm/hr aside from the load which was run at 112 cm/hr for the MabSelect SuRe column and 150 cm/hr for the MabSelect Prisma column to ensure typical process load residence times.

Table 3.1-1: Dynamic Binding Capacity Study Conditions

Run	Resin	1.4% Minncare Exposure (Hours)	Load Concentration (g/L)
1	MabSelect SuRe	0	5.04
2	MabSelect SuRe	1	5.04
3	MabSelect SuRe	2	5.04
4	MabSelect SuRe	24	5.04
5	MabSelect SuRe	70	5.04
6	MabSelect SuRe	140	4.77
7	MabSelect Prisma	0	2.68
8	MabSelect Prisma	144	2.68

3.2 Product Quality Comparison Study Methods

The product used for the product quality comparison study was MEDI9447 CM. The cell culture material was 0.2 micron filtered and brought to room temperature before use. The same 1.1 x 9.9 cm column packed with MabSelect Prisma was used for this study. Product quality was taken before and after the exposures mentioned above. For Minncare exposure the column was equilibrated with 3CV of 1.4% Minncare solution and held for 140 hours wrapped in foil to prevent light exposure. The column was run in the following conditions for

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both pre and post Minncare exposure, which is in accordance to the MEDI9447 process description across MabSelect SuRe:

- 1) Equilibration with 3CV 50 mM Tris, pH 7.4
- 2) Load MEDI9447 CM
- 3) Equilibration 2 with 3CV 50 mM Tris, pH 7.4
- 4) Wash with 3CV 100 mM tris, 50 mM sodium caprylate, 1 M sodium chloride, pH 9.0
- 5) Equilibration 3 with 3CV 50 mM Tris, pH 7.4
- 6) Elution with 25 mM sodium acetate, pH 3.6, starting and stopping collection at 0.5OD
- 7) Strip with 2CV 100 mM Acetic Acid
- 8) Caustic Strip with 3CV 0.1N Sodium Hydroxide with 30 minute hold
- 9) Storage with 3CV 2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0

All steps were run at 300 cm/hr aside from the load which was run at 150 cm/hr to achieve a 4 minute residence time.

3.3 Residual Minncare Methods

Residual Minncare was tested for by equilibrating the MabSelect SuRe column in 1.4% Minncare solution and chasing with MEDI column storage solution; 2% Benzyl Alcohol, 100 mM Sodium Acetate, pH 5.0. Minncare Residual Test Strips were used to determine how much Minncare remained in the eluate. The test strips are color indicative with six color reaction zones correlating to 100, 30, 10, 3, 1 and 0 ppm Minncare.

3.4 1.4% Minncare Expiry Methods

The ability of Minncare to effectively kill endospores lies in its hydrogen peroxide and peracetic acid content. Hydrogen peroxide is light sensitive, so it is important to determine how long Minncare can be stored before it is no longer effective. Two 1.4% Minncare solutions were made and stored under a florescent light in a fume hood at room temperature (15-25°C) and tested for hydrogen peroxide concentration every day up to 7 days using Minncare 1% test strips. The test strips are color indicative, turning dark blue or black if the

Minncare concentration is at or above 1%, and remaining white or light blue if the concentration is under 1%.

4 RESULTS AND DISCUSSION

4.1 Dynamic Binding Capacity Studies

The study was performed without any unplanned events. The breakthrough curves for each run are shown in Figure 4.1-1. It can be seen that the curve shape stays uniform for every level of exposure with the minor exception of the 140-hour exposure, which appears to breakthrough slightly quicker, but only after 10% breakthrough, which is typically used as a development benchmark.

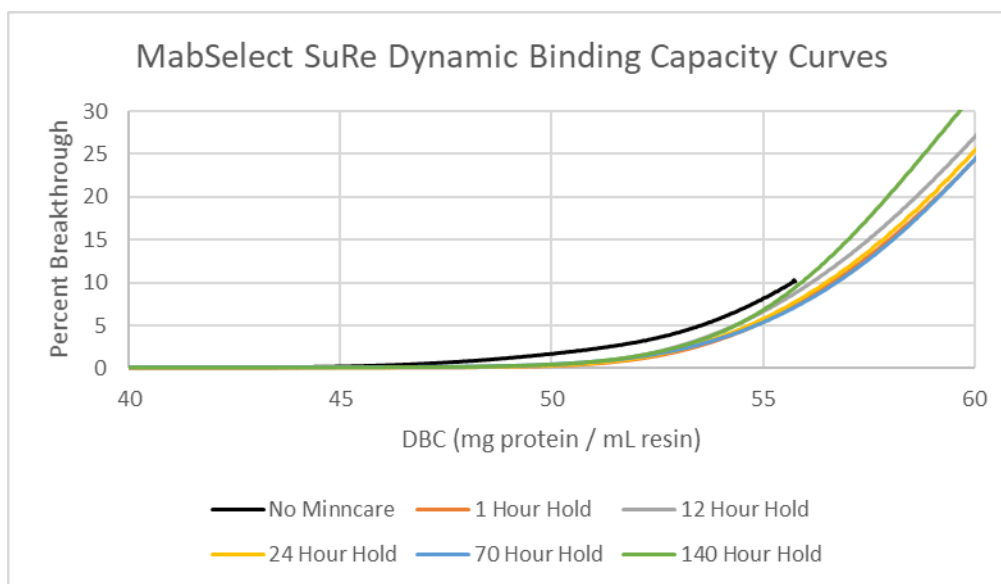


Figure 4.1-1. Chromatograms of MEDI4893 breakthrough on MabSelect Sure resin after 0, 1, 12, 24, 70, and 140 hours of static 1.4% Minncare solution exposure to the resin.

The same experiment repeated with MabSelect Prisma revealed similar results with chromatograms shown in Figure 4.1-2. Again, the dynamic binding capacity curves show very similar characteristics showing that Minncare 1.4% solution does not degrade the ligand on MabSelect Prisma.

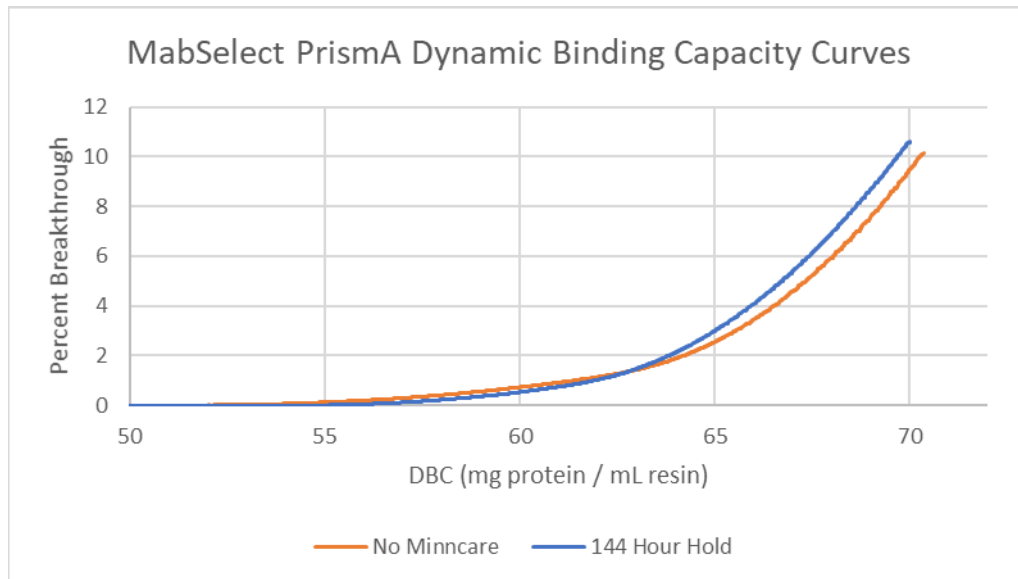


Figure 4.1-2. Chromatograms of MEDI4893 breakthrough on MabSelect PrismaA resin after 0 and 144 hours of static 1.4% Minncare solution exposure to the resin.

The 10% breakthrough binding capacities for all runs are shown in Table 4.1-1. This demonstrates that Minncare 1.4% does not degrade the protein A ligand significantly enough to be seen through a reduction in protein binding capacity when held statically up to 140 hours of exposure.

Run	Resin	Minncare Exposure (Hours)	DBC at 10% BT (mg/mL)
1	MabSelect SuRe	0	55.7
2	MabSelect SuRe	1	56.0
3	MabSelect SuRe	2	55.8
4	MabSelect SuRe	24	55.8
5	MabSelect SuRe	70	55.8
6	MabSelect SuRe	140	55.9
7	MabSelect PrismaA	0	70.2
8	MabSelect PrismaA	144	69.7

Table 4.1-1. Dynamic binding capacity (DBC) at 10% breakthrough for MEDI4893 on MabSelect SuRe, after 0, 1, 12, 24, 70, and 140 hours of 1.4% Minncare solution exposure (Runs 1-6), and MabSelect PrismaA after 0 and 144 hours of 1.4% Minncare solution exposure (Runs 7 and 8).

4.2 Product Quality Results

The study was performed without any unplanned events. The product quality attributes of the eluate tested were protein concentration, DNA, purity, residual protein A, insulin, silicone, Pluronic F68, charge variants, host cell protein, and yield. The results of these measurements are displayed in Table 4.2-1.

The results show a minor decrease in protein concentration, but with no loss of monomer yield as the elution volume increased proportionally. This again, indicates Minncare does not impact the resin ligand's ability to bind to protein. There was a noticeable increase in DNA and host cell protein levels and a relatively large increase in residual protein A. While this may suggest there is some impact to product quality at 140 hours of static exposure, it is worth noting that these levels are within typical process limits and can normally be cleared by subsequent downstream unit operations. More data must be collected to confirm these results.

Table 4.2-1. Product quality results for MEDI9447 across MabSelect Prisma before and after 140 hours of 1.4% Minncare solution exposure to the resin. An asterisk (*) indicates the results are at or below the lower limit of quantification of the assay.

Product Quality Attribute	No Minncare Exposure	140 Hours Minncare Exposure
Eluate Concentration (mg/mL)	15.76	14.02
Elution Volume (CV)	2.4	2.7
Step Yield (%)	87.8	88.1
DNA (ng/mg)	0.09	0.16
Purity (% mon, %agg, %frag)	94.6, 2.2, 3.3	93.8, 2.6, 3.6
Residual Protein A (ng/mg)	3.5*	18.3
Silicone (Si ppm, Simeth ppm)	<0.5*, <1.32*	<0.5*, <1.32*
Pluronic F68 (µg/mL)	None Detected	None Detected
Charge Variants (%Main Peak, %Acidic, %Basic)	44.9, 51.8, 3.4	43.8, 52.8, 3.4
Host Cell Protein (ng/mg)	2732	3535

4.3 Residual Minncare Results

The study was performed without any unplanned events. The results are summarized in Table 4.3-1. It is shown that it takes 7 CV of storage solution to reduce the Minncare levels below 1 ppm.

Flush Volume (CV)	Residual Minncare (ppm)
3	3-10*
4	3
5	3
7	1

Table 4. Residual Minncare in eluate after flushing with 3, 4, 5, and 7 column volumes of 2% Benzyl Alcohol, 100 mM Sodium Acetate, pH 5.0. *Test strips were color based and a color between two reference colors was reported as a range between the two.

4.4 Minncare Storage Results

The study was performed without any unplanned events. The test strips revealed no detected decrease in concentration for all 7 days indicating that the 1.4% solution is light stable, in that it does not fall below 1.0% concentration, for at least a week when stored at room temperature.

5 RECOMMENDATIONS

5.1 Minncare Usage

Static Minncare exposure has shown to have no impact on dynamic binding capacity at 10% breakthrough for MabSelect SuRe and MabSelect PrismA, but has shown to have some negative effects on product quality with regards to DNA, HCP, and especially residual protein A. However, these effects were observed at an exposure time of 140 hours which was determined to be the worst-case scenario. This scenario assumed a 2-hour static exposure time run every 3-cycle extended out to a 200-cycle resin lifetime. The 2-hour exposure time was based on spore kill time data from NG-1861358 (REPT_ANAL(370477)) shown in Appendix, Table 3. In practice it would be recommended to only use Minncare as needed, which would be when bioburden data shows normal sanitizing procedures did not kill all spores. After exposure it is recommended to flush with at least 7 CV of 2% Benzyl Alcohol 50mM Sodium Acetate, pH 5.0 based on data collected thus far. Other storage or

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intermediate buffers may be more effective at flushing Minncare and should be investigated to possibly reduce the high 7CV flush volume.

5.2 Minncare Storage

Minncare uses hydrogen peroxide as one of its active ingredients which means it is susceptible to light and temperature. It is also classified as an oxidizer and should be stored in a designated oxidizer storage compartment away from organics and flammable materials. Stock Minncare solution should be stored in the dark and not exposed to heat based on vendor recommendations. The 1.4% Minncare solution however does not appear to be as sensitive to light degradation, meaning the solution should be able to be stored in typical polyethylene buffer bags for up to at least 7 days. Mar Cor technical document 50096-318 also states up to full stock strength Minncare is compatible with polyethylene.

5.3 Future Work

The next study to be performed is a lifetime study which will demonstrate how dynamic binding capacity and product quality change over time with dynamic Minncare exposure. Dynamic exposure will subject the resin to two-hour increments of fresh 1.4% Minncare solution up to 140 hours, rather than 140 hours at a time. This will realistically measure how Minncare will affect the resin and product quality in practice.

6 CONCLUSIONS

Minncare can be effectively used to kill endospores on columns otherwise unaffected by traditional 0.1 N hydroxide exposure (Appendix). Static holds up to 140 hours demonstrate there is little impact on breakthrough curve characteristics and no impact on dynamic binding capacity at 10% breakthrough when used on MabSelect SuRe and MabSelect Prisma. Yield was unaffected after 140 hours of 1.4% Minncare exposure on MabSelect Prisma and product quality was mostly unchanged with the exceptions of DNA, HCP, and residual protein A which all increased, especially residual protein A with an increase of over 4x. Overall, this study shows that 1.4% Minncare is a promising endospore disinfectant for MabSelect SuRe and MabSelect Prisma resins, potentially addressing an unmet need in commonly used Protein A resin sanitization buffers. Further research is warranted to fully elucidate the impact to product quality and process performance throughout resin lifetime.

7 REFERENCES

Compatibility of Minncare® Cold Sterilant and System Materials (P/N 50096-318)

NG-1861358 (REPT_ANAL(370477)) PAB vs. Minncare vs. 0.1N NaOH

eLN ML02604-04

8 APPENDIX

MedImmune originated *Bacillus* and *A. Brasiliensis* (16404) were separately spiked in the presence of MabSelect SuRe and exposed to NaOH, PAB, or 1.4% Minncare. Log recovery data was evaluated by culture forming units sampled and quenched/diluted with Tryptone-Azolectin-Tween (TAT) Broth at 0, 8, 14, 28, 55, 115, and 235 minutes. The MedImmune originated *Bacillus* was cultured on Tryptic Soy agar, 100mm dishes and the *A. Brasiliensis* was cultured on Sabouraud Dextrose agar, 100mm Dishes. Microbial recovery method validation was performed according to “Microbial Challenge Efficacy Study of Select Chromatography Resins, MedImmune” (1-P-QM-WI-9024066) drafted by Eurofins Lancaster Laboratories Inc.

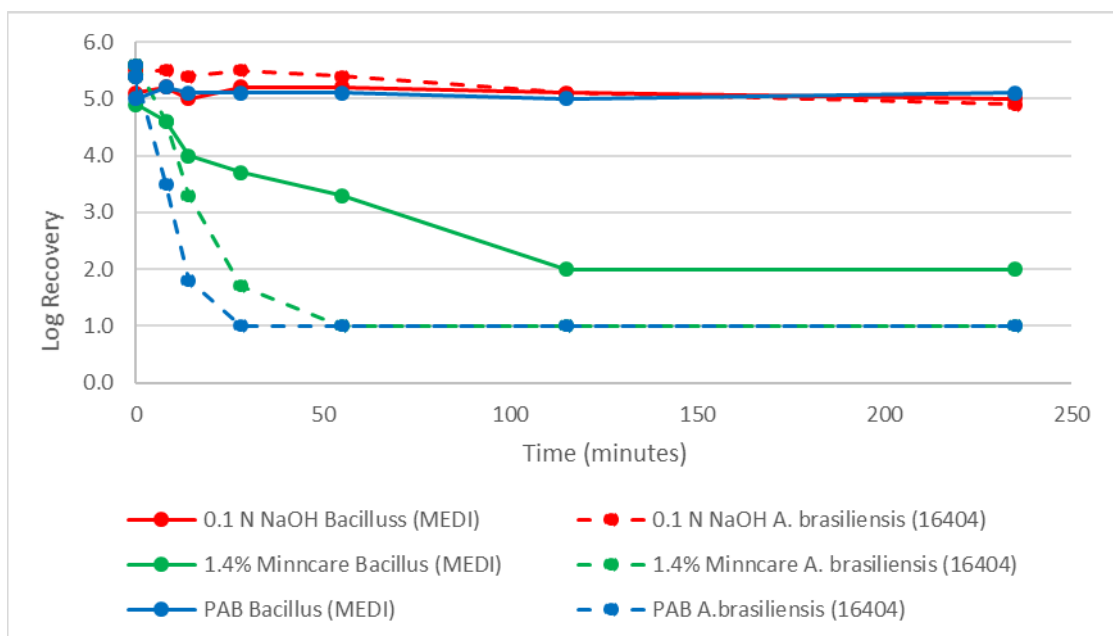


Figure 3. Log reduction values of MedImmune originated *Bacillus* (solid lines) and *A. brasiliensis* (16404) (dashed lines) on a MabSelect SuRe column equilibrated in 0.1N Sodium Hydroxide (red), 1.4% Minncare Solution (green), and PAB (blue). Data taken from NG-1861358 (REPT_ANAL(370477)).

9 EDITION HISTORY

Edition	Reason
001	New document